

# Networks

## Definitions

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Optimization: principles and algorithms



# Definitions

*Networks are everywhere.*

*We introduce a mathematical formalism that mimics the structure of real networks.*

# Road networks



## Public transportation networks



# Electricity networks



# Gas networks



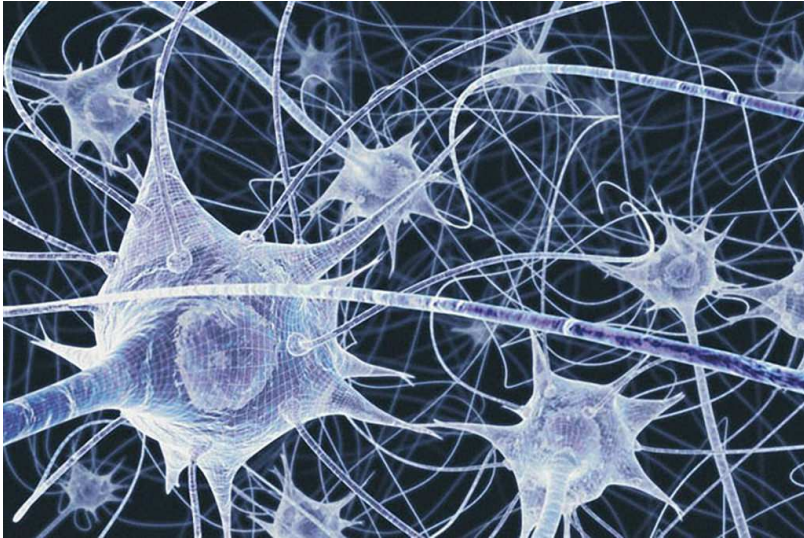
# Water networks



# Computer networks



# Neural networks



# Social networks



# Networks

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System of interconnected people or things.

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## Main features

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## Mathematical object

Similar property: designed to capture complex structures with simple elements.

# Graph

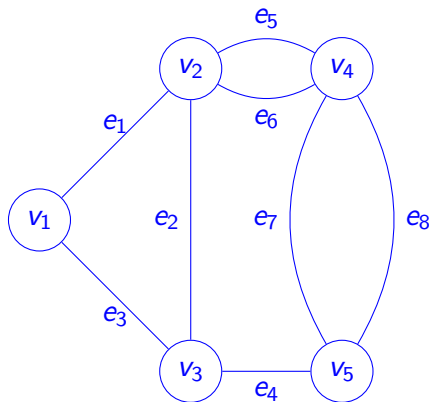
## Definition

- ▶  $\mathcal{V}$ : set of vertices.
- ▶  $\mathcal{E}$ : set of edges.
- ▶  $\phi : \mathcal{E} \rightarrow \mathcal{P}_2(\mathcal{V})$ : incidence function.

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# Graph

## Example

$$\mathcal{V} = \{v_1, v_2, v_3, v_4, v_5\},$$

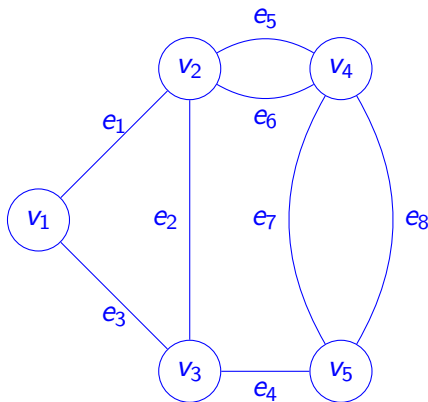
$$\mathcal{E} = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$$

$$\phi(e_1) = \{v_1, v_2\}, \phi(e_2) = \{v_2, v_3\},$$

$$\phi(e_3) = \{v_1, v_3\}, \phi(e_4) = \{v_3, v_5\},$$

$$\phi(e_5) = \{v_2, v_4\}, \phi(e_6) = \{v_2, v_4\},$$

$$\phi(e_7) = \{v_4, v_5\}, \phi(e_8) = \{v_4, v_5\}.$$

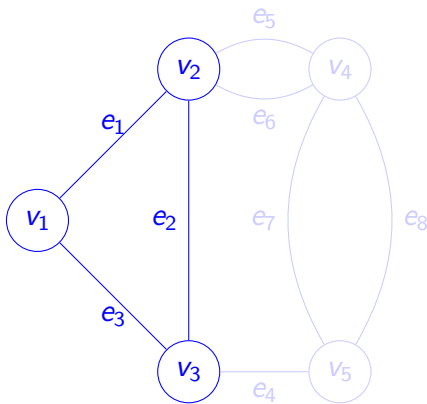


# Subgraph

## Definition

$(\mathcal{V}', \mathcal{E}', \phi')$  is a subgraph of  $(\mathcal{V}, \mathcal{E}, \phi)$  if

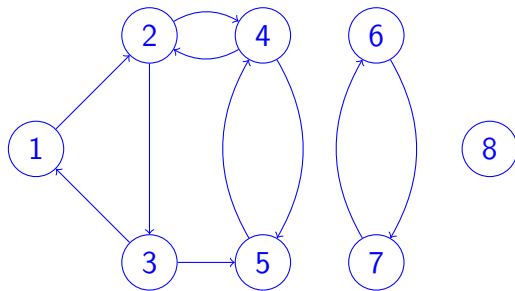
- ▶  $\mathcal{V}' \subseteq \mathcal{V}$ ,
- ▶  $\mathcal{E}' \subseteq \mathcal{E}$ ,
- ▶  $\phi'(e) = \phi(e)$ , for each  $e \in \mathcal{E}'$ ,
- ▶ for each  $e \in \mathcal{E}'$ , if  $\phi'(e) = \{i, j\}$ , then  $i$  and  $j$  both belong to  $\mathcal{V}'$ .



# Directed graph

## Definition

- ▶  $\mathcal{N}$ : set of nodes.
- ▶  $\mathcal{A}$ : set of arcs.
- ▶  $\phi : \mathcal{A} \rightarrow \mathcal{N} \times \mathcal{N}$ : incidence function.



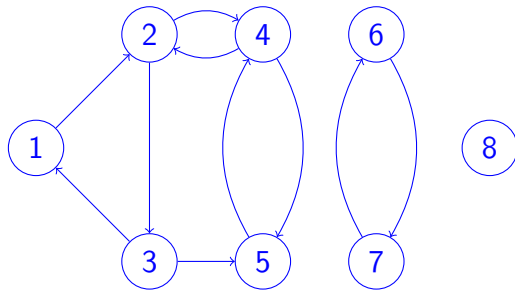
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## Assumption

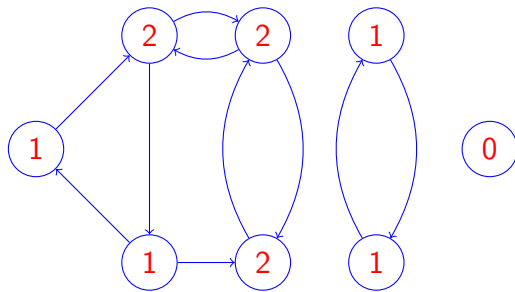
- ▶  $\phi$  is injective.
- ▶ For each  $(i,j)$ , there is at most one  $a$  such that  $\phi(a) = (i,j)$ .
- ▶ Arcs are denoted by  $(i,j)$ .



# Indegree

## Definition

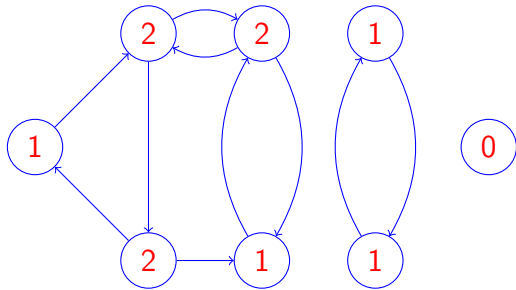
$d_i^-$ : number of arcs  $(j, i)$ .



# Outdegree

## Definition

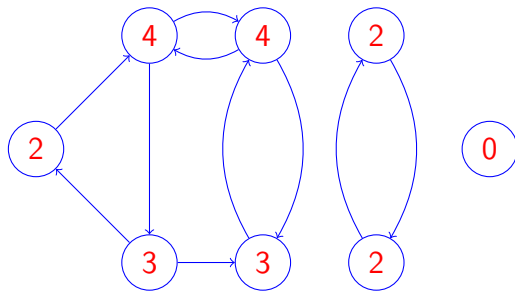
$d_i^+$ : number of arcs  $(i, j)$ .



# Degree

## Definition

$$d_i = d_i^- + d_i^+.$$



# Summary

*Graphs capture the relationships among objects.*

*The objects are represented by vertices.*

*Their relationship by edges.*

*When the relationship has a direction, we talk about directed graphs.*

*The vertices are called nodes.*

*The edges are called arcs.*